

Article

Cost Allocation Methods and Their Properties in Energy Communities

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Abstract

Energy communities enable prosumers to jointly operate distributed energy resources and thereby generate economic benefits that exceed those achievable individually. A central challenge in their implementation is selecting a Cost Allocation Method (CAM) that distributes these benefits fairly among heterogeneous participants. Although numerous CAMs have been proposed, they are often evaluated under different assumptions, making direct comparison difficult. This paper develops a unified axiomatic framework for assessing CAMs in energy communities and applies it to eight representative methods classified in three families: simple rules, savings-based, and price-based. The framework is built around seven desirable properties capturing principles of fairness, environmental friendliness, and continuity. Our main contribution is a comparative table that positions all methods within a single evaluative space and reveals the structural trade-offs that arise across CAMs. The analysis shows that the Average-Price CAM satisfies the same axiomatic properties as the Shapley method while remaining computationally trivial, making it an attractive practical option. We also show that the Extreme-Price CAM is the only price-based method that ensures the property of Beneficial Group Participation (core stability); however, this method violates other properties related to environmental friendliness and continuity—trade-offs we prove to be unavoidable for price-based rules. Finally, we conjecture that the nucleolus satisfies all seven properties, although its computation is rarely feasible in practice. The proposed framework provides researchers and practitioners with a transparent foundation for selecting and designing cost allocation methods in emerging energy communities.

Keywords: energy communities; cost allocation; fairness; nucleolus; Shapley value

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1. Introduction

The global energy system is undergoing a profound transformation driven by the dual imperatives of decarbonization and decentralization. Increasing shares of renewable generation, much of it distributed and non-dispatchable, are reshaping the traditional top-down architecture of power systems into a more decentralized, consumer-centric structure [1]. In this new landscape, prosumers—agents that both consume and produce energy—are emerging as pivotal actors, providing operational flexibility and contributing to the integration of variable renewable resources [2,3]. Policy frameworks across the European Union [4,5] and other regions have recognized the potential of prosumers, encouraging their active involvement through the creation of Energy Communities (ECs), which enable

groups of citizens, households, or small businesses to jointly invest in, produce, consume, and manage energy.

Energy Communities have the potential to improve system efficiency, enhance social acceptance of renewables, and deliver economic and environmental benefits to their members [6]. In general, energy sharing results in cost savings for the community because the price that members must pay to purchase one unit of energy from the grid is generally strictly higher than the price they receive when they sell energy to the grid [7,8]. Thus, if they share energy among themselves, they save this price spread [9]. The natural question is then: How are these aggregate savings shared among the members of the community?

Realizing the full potential of ECs requires the design of appropriate internal rules to govern how costs and benefits are allocated among participants. Because members differ in consumption profiles, renewable production, flexibility, and asset ownership, the selection of a Cost Allocation Method (CAM) is central to fairness perceptions, participation incentives, and long-term stability. If the allocation method is not perceived as beneficial, fair, and adequate by the EC members, the long-term sustainability of the EC is jeopardized. This challenge is well recognized in the broader literature on cooperative energy management, where the design of appropriate rules to allocate costs and benefits has emerged as a central research theme [1,2,7,10–17].

Despite a rapidly expanding literature, to the best of our knowledge, there is currently no unified framework that allows researchers, practitioners, and policymakers to compare existing CAMs on a common basis. Most studies analyze a subset of methods, often using different evaluative criteria or different case-study conditions, which makes it difficult to understand the overarching structure of the design space and the fundamental trade-offs between fairness, incentives, and computational feasibility.

This paper addresses this gap by providing a unified axiomatic characterization of cost allocation methods for energy communities.

We identify a set of core properties that capture principles of fairness, environmental friendliness, and continuity, and we apply these properties to systematically evaluate eight representative CAMs commonly used in the literature. Our main contribution is a comprehensive comparison table that places all methods within the same evaluative space, thereby making explicit the trade-offs that arise across different approaches.

The analysis reveals that no computationally efficient CAM satisfies all desirable properties simultaneously, highlighting structural tensions between fairness axioms, incentive compatibility, and ease of implementation. Among our findings, we show that the Average-Price CAM satisfies the same axiomatic properties as the Shapley method while remaining computationally trivial. The Extreme-Price CAM is the only price-based method that ensures beneficial group participation (core stability), though at the cost of violating other desiderata. Finally, we conjecture that the nucleolus satisfies all desirable properties, but its high computational complexity makes it impractical for real-world applications. By offering a unified lens through which CAMs can be evaluated, the paper provides a foundation for designing and selecting cost allocation methods in emerging energy communities.

1.1. Related Work

Research on cost allocation in energy communities spans several complementary lines, ranging from fairness formulations and empirical assessments to cooperative-game approaches and multiagent simulations.

A first line of work examines how different fairness criteria shape the allocation of costs and benefits in ECs. Volpato et al. [10] analyze the distribution of economic benefits by comparing cooperative-game-theoretic and optimization-based criteria under alternative

fairness paradigms. Applying a social-welfare maximization model and a Nash-bargaining framework to a small-scale EC, they redistribute aggregate gains using the Shapley value and the nucleolus and find substantial variation in outcomes depending on the fairness notion adopted. Their results show that “price of fairness” trade-offs can reach 5.5% of total savings while favoring lower-demand consumers, highlighting that no single method satisfies all reasonable fairness criteria and that explicit property-based comparison is essential.

Empirical contributions reinforce this need for systematic evaluation. Limmer [11] provides an empirical assessment of benefit-distribution methods in local energy communities using real household data, comparing three allocation methods: the Shapley value, the nucleolus, and the Shapley-core. The study evaluates these methods using two criteria: fairness and stability. Fairness is defined by taking the Shapley value as the benchmark, following the author’s assumption that the Shapley allocation represents the fairest distribution. This yields a clear and operational fairness criterion, although one that is tied to a specific allocation rule rather than to a general set of fairness properties. Second, stability is defined using the cooperative-game-theoretic concept of the core, whereby an allocation is considered stable if no coalition can improve its benefit by leaving the community. This choice—using Shapley to define fairness and the core for stability—illustrates how different studies adopt different formalizations of key evaluative notions. Empirically, Limmer finds that the Shapley value often performs well in smaller communities, while the nucleolus and the Shapley-core become more advantageous as community size increases, albeit with significantly higher computational demands. Taken together, the results show how conclusions about benefit distribution depend on which formal principles are chosen to represent fairness and stability.

Another important strand of research examines community formation and operational efficiency. Volpato et al. [12] investigate how the economic performance of energy communities depends on prosumer complementarity, demand-response options, and the choice of cost allocation method. Using an optimization framework, they show that complementary consumption–generation profiles and demand-response programs can substantially reduce community costs, while different allocation methods lead to markedly different benefit distributions. Their results illustrate how operational features and allocation choices jointly shape the internal economics of energy communities.

A comprehensive perspective is provided by Li, Hakvoort, and Lukszo [13–15], who examine cost allocation in Integrated Community Energy Systems from regulatory, social, and operational angles. Their review [13] identifies the main families of cost allocation methods and the principles underlying them—such as simplicity, transparency, cost causality, and non-discrimination—and shows how these concepts guide tariff design in traditional electricity systems. In a companion paper on social acceptance [14], the authors investigate how community members assess fairness in practice and demonstrate that perceptions of fairness depend strongly on procedural transparency and the understandability of the method; in particular, they show that a method that is normatively fair according to regulatory or economic principles may not be perceived as fair by users if it is difficult to explain or conflicts with community expectations. Their performance assessment [15] further evaluates allocation methods using operational criteria such as cost reflectiveness and cost predictability, highlighting additional trade-offs that arise when methods are applied in realistic settings. Collectively, these contributions show that fairness in energy communities has multiple dimensions—formal, behavioral, and operational—and that the terminology used to describe these dimensions is not always applied consistently across the literature.

Additional studies analyze energy sharing through multi-agent and coalition-formation approaches. Zhou, Wu, and Long [16] use agent-based simulations to compare

peer-to-peer energy-sharing methods and show that both efficiency and distributional outcomes depend sensitively on behavioral assumptions and market rules. Cooperative-game-theoretic analyses by Safdarian et al. [17] and Feng et al. [7] examine cost sharing in coalitional settings and highlight the appeal of methods that satisfy individual rationality and core stability, although these models often involve substantial computational effort and can be difficult to interpret in practical community contexts. Together, these studies illustrate how conclusions about fairness and stability can vary widely depending on modeling choices.

A recurring challenge in this literature is that terms such as “fairness,” “stability,” and “equity” are used with different meanings in different papers, and are sometimes invoked without a precise formal definition. As a result, methods are frequently compared using heterogeneous or implicit criteria, making it difficult to determine which differences arise from fundamental theoretical properties and which stem from contextual assumptions or implementation choices. This lack of a shared formal vocabulary limits comparability and hinders cumulative progress in the field.

In contrast, the present paper formalizes all properties explicitly within an axiomatic framework, ensuring that fairness-related concepts are defined unambiguously and evaluated consistently across methods. This eliminates potential misunderstandings and enables transparent, like-for-like comparison of cost allocation methods, thereby addressing a key methodological gap in the existing literature.

1.2. Structure of This Paper

The remainder of the paper is organized as follows. Section 2 presents the materials and methods used in the analysis. It introduces the setting of an energy community with internal energy sharing, defines the notions of net consumers and net producers, and explains the eight cost allocation methods (CAMs) examined in the study. It then introduces the seven desirable axiomatic properties used to evaluate these methods and explains the fully analytical procedure employed to assess property compliance.

Section 3 reports and discusses the results. It summarizes, for each CAM, which properties are fulfilled and highlights the key trade-offs that emerge across simple-rule, savings-based, and price-based approaches. Section 4 concludes the paper by synthesizing the main findings, discussing their implications for the design of cost allocation rules in energy communities, and outlining avenues for future research.

All formal proofs supporting the analytical results are collected in Appendix A, while Appendix B contains the proofs of propositions presented in the main text.

2. Materials and Methods

This section introduces the setting of the analysis, defines the cost allocation methods (CAMs) considered, and presents the seven axiomatic properties used to evaluate them within a unified framework. The section ends explaining the approach followed to check which properties are fulfilled by each CAM, which is fully analytical. Rather than relying on simulations, all results are derived through formal reasoning, allowing us to determine property compliance under any consumption–generation configuration.

2.1. Total Cost for an EC That Can Share Energy Internally

In this section, we compute the total cost incurred by an energy community (EC) whose members are allowed to share energy among themselves. We use a stylized EC model that captures the essential structure shared across many contributions in the literature and allows us to examine CAMs independently of operational or regulatory details. For

convenience, all symbols used in this section are summarized in the Symbols list at the end of the manuscript.

Let A denote the energy community and let a represent an individual member (or agent) of the community. The price of purchasing energy from the grid is denoted by p_b , and the price at which energy can be sold to the grid is denoted by p_s . We assume that $p_b > p_s > 0$ and there is no marginal cost for obtaining energy from the shared generation unit. All variables refer to a specific time slot t , although we omit the time index throughout for notational simplicity. A time slot (or compensation period) is the interval over which the energy consumed by the community can be directly compensated with energy generated within the community. In grid-connected communities, the maximum duration of a time slot is typically set by regulation.

Let E^g denote the energy generated by the community and E^c the total energy consumed. The total cost incurred by a community that allows internal energy sharing is then given by:

$$C = p_b[E^c - E^g]^+ - p_s[E^g - E^c]^+ \quad (1)$$

where $[x]^+ = \max(0, x)$. Observe that when the community's consumption exceeds its generation ($E^c > E^g$), the expression $[E^g - E^c]^+$ evaluates to zero. Conversely, when generation exceeds consumption ($E^g > E^c$), we have $[E^c - E^g]^+ = 0$.

The EC model used here is intentionally minimal, reflecting the most common structure in the literature. It assumes known and deterministic consumption and generation within each time slot, fixed grid prices, and no marginal cost of internal transfers. This abstraction allows us to focus purely on the theoretical properties of ex-post CAMs (those which distribute savings *after* the time slot has ended), without conflating them with operational or regulatory variations. More complex settings like dynamic pricing, storage, or network constraints can be analyzed within the same axiomatic framework, but are beyond the scope of this paper.

In this paper we examine different ex-post methods that can be used to allocate the total net cost C (which may be negative) among the members of the community. To conduct this allocation, it is often useful to distinguish between net consumers and net producers in the community. These concepts are explained in the next section.

2.2. Definition of Net Consumers and Net Producers

Most ECs distribute the energy produced by the shared generation unit among its members according to some pre-established criterion. Two common ways of conducting this allocation are (a) to assign every member an equal proportion of the generated energy E^g , and (b) to allocate E^g proportional to the members' investment on the energy-generating unit.

We allow for an arbitrary allocation of the generated energy E^g by introducing α_a , the fraction of E^g assigned to agent $a \in A$. We make no specific assumptions about the values of these fractions α_a other than they should add up to 1. The energy obtained by agent a from the generation unit is denoted by $e_a^g = \alpha_a E^g$, while we use e_a^c to denote the energy consumed by agent a . Thus, we have $\sum_{a \in A} e_a^g = E^g$ and $\sum_{a \in A} e_a^c = E^c$.

We can now define the set of Net Consumers (NC) and the set of Net Producers (NP) in a certain time slot t . Net Consumers are those agents who consume more than their allocated energy from the generation unit (i.e., $e_a^c > e_a^g$). Net Producers are those agents who consume less than their allocated energy from the unit (i.e., $e_a^c < e_a^g$). We also define the set $N0$ as the set of agents who consume exactly their allocated energy (i.e., $e_a^c = e_a^g$).

We define the overconsumption e_a^{NC} of agent $a \in A$ as $e_a^{NC} = [e_a^c - e_a^g]^+$. Thus, for a Net Consumer $a \in NC$, $e_a^{NC} = (e_a^c - e_a^g)$ is the part of a 's consumption that is

not satisfied by the energy obtained from the generation unit, while for a net producer, $e_{a \in NP}^{NC} = 0$. Similarly, we define the overproduction e_a^{NP} of agent $a \in A$ as $e_a^{NP} = [e_a^g - e_a^c]^+$. Thus, for a Net Producer $a \in NP$, $e_{a \in NP}^{NP} = (e_a^g - e_a^c)$ is her energy excess, while for a net consumer, $e_{a \in NC}^{NP} = 0$.

If sharing energy within the community was not allowed, the cost for an individual prosumer a would be

$$c_a^0 = p_b e_a^{NC} - p_s e_a^{NP} \quad (2)$$

In that case, NC as a group would have to buy $E^{NC} = \sum_{a \in NC} e_a^{NC}$ units of energy from the grid, and NP as a group would sell all their energy excess $E^{NP} = \sum_{a \in NP} e_a^{NP}$ to the grid. Thus, the total cost for the community would be

$$C^0 = \sum_{a \in A} c_a^0 = p_b E^{NC} - p_s E^{NP} \quad (3)$$

However, when sharing is allowed, Net Producers can transfer their energy excess to Net Consumers and, given that $p_b > p_s$, the community as a whole can save money. Every unit that is transferred from a net producer to a net consumer implies a saving for the community of the price spread $(p_b - p_s)$. To maximize the savings, net producers can transfer to net consumers the amount $E^{Tr} = \min(E^{NC}, E^{NP})$ and, in that case, the total savings achieved by the community equal $(p_b - p_s)E^{Tr}$. Thus, the total cost for an EC that can share energy internally (see Equation (1)) can also be written as:

$$C = C^0 - (p_b - p_s)E^{Tr} = p_b E^{NC} - p_s E^{NP} - (p_b - p_s)E^{Tr} \quad (4)$$

2.3. Cost Allocation Methods

A Cost Allocation Method (CAM) is an algorithm that distributes the net total cost C among the members of the EC. In this section we present eight CAMs that are widely used or frequently proposed in the EC literature and reflect the full range of fairness and incentive properties discussed in prior research. We classify the different allocation methods into three families: simple rules, savings-based, and price-based allocation methods. The following sections explain these families in detail.

2.3.1. Family of Simple Rules

The family of simple rules comprises two rules that are easy to apply and understand: the All Equal rule and the Bill Sharing rule.

- **All Equal**

The All Equal (AE) rule distributes the total net cost equally among the n participants, regardless of their individual energy consumption or contribution. Formally, the allocated cost for each participant $a \in A$ under this CAM is given by:

$$c_a^{AE} = \frac{C}{n} \quad (5)$$

- **Bill Sharing**

Under the Bill-Sharing (BS) rule [16,18,19], if the total energy produced by the generation unit exceeds the total consumption of the community ($E^g > E^c$), then the excess is sold to the grid, and the income obtained is shared among the members proportional to their overproduction e_a^{NP} (so only Net Producers obtain some income). On the other hand, if the community needs to buy energy from the grid ($E^c > E^g$), then the cost of this bill is distributed among the members proportional to their overconsumption e_a^{NC} (so only Net Consumers pay the bill).

This is the so-called bill sharing method net, put forward by Grzanic et al. [19] as an upgrade to the bill sharing method proposed by Long et al. [18]. Formally, the allocated cost for member $a \in A$ under this CAM is given by:

$$c_a^{BS} = \begin{cases} -\frac{e_a^{NP}}{\sum_i e_i^{NP}} p_s (E^g - E^c) & \text{If } E^g \geq E^c \\ \frac{e_a^{NC}}{\sum_i e_i^{NC}} p_b (E^c - E^g) & \text{If } E^g < E^c \end{cases} \quad (6)$$

2.3.2. Family of Savings-Based Allocation Methods

In Section 2.2 we saw that allowing energy sharing among members of a community A leads to community-wide savings equal to $(p_b - p_s)E^{Tr}$. The family of savings-based allocation methods distributes these savings according to some rule. Here, we discuss four specific instances of the family of savings-based allocation systems: two straightforward methods (distribution of savings proportional to pre-established shares β_a , and distribution of savings proportional to coefficients α_a), and two game-theoretic methods that have played a prominent role in this line of research [7,10,17,20], namely the Shapley CAM and the nucleolus CAM.

Both the Shapley CAM and the nucleolus CAM require the definition of a coalitional game with transferrable payoff [21] (part IV), which is defined by a set of players and a characteristic function. In our case, the set of players is A , i.e., the members of the community. The characteristic function $v(S)$ assigns a value to any subset, or coalition, $S \subseteq A$. In our case, the value $v(S)$ is the total savings that the subset S of community members would obtain by internal sharing if they decided to part ways and form their own community. Thus, $v(S) = (p_b - p_s) \min(\sum_{a \in S} e_a^{NC}, \sum_{a \in S} e_a^{NP})$. The game thus defined is cohesive [21] (part IV) and, consequently, superadditive and monotonic.

Both the Shapley CAM and the nucleolus CAM have been proposed as allocation rules that balance fairness and stability in cooperative energy management [1,3]. The Shapley value is appealing for its axiomatic characterization and intuitive interpretation in terms of marginal contributions, while the nucleolus minimizes the maximum dissatisfaction across all coalitions, offering strong stability guarantees. Yet, despite their desirable properties, these methods are computationally demanding in practice, so their implementation in real-world energy communities remains limited.

- Distribution of savings proportional to pre-established shares

This allocation method distributes the savings generated by internal sharing proportional to pre-established shares β_a (potentially different for each agent). These shares may also be different in different time slots, but are fixed for each agent within each time slot. Without loss of generality, we assume that shares are normalized so $\sum_{a \in A} \beta_a = 1$. Therefore, the individual allocated cost for member a under this CAM is:

$$c_a^\beta = c_a^0 - \beta_a (p_b - p_s) E^{Tr} \quad (7)$$

- Distribution of savings proportional to coefficients α_a

This allocation method distributes the savings generated by internal sharing proportional to coefficients α_a , i.e., the fraction of generated energy assigned to each agent. Naturally, this method is a particular case of the previous CAM where $\beta_a = \alpha_a$ for all $a \in A$. Thus:

$$c_a^\alpha = c_a^0 - \alpha_a (p_b - p_s) E^{Tr}$$

- Distribution of savings using the Shapley value

The Shapley CAM requires the definition of a coalitional game with transferrable payoff (A, v) , as introduced at the beginning of this Section 2.3.2. The Shapley value [22] of this game provides a method for distributing the total savings $v(A) = (p_b - p_s)E^{Tr}$ among the n members of the community A . It allocates to each member a weighted average of her marginal contributions over the set of all possible coalitions that can be formed. Formally, given the coalitional game (A, v) , the Shapley value for agent $a \in A$, v_a^{Sh} , is:

$$v_a^{Sh} = \sum_{S \subseteq A \setminus \{a\}} \frac{|S|!(n - |S| - 1)!}{n!} (v(S \cup \{a\}) - v(S)) \quad (8)$$

The Shapley value is the unique solution that satisfies four fundamental properties: efficiency, symmetry, additivity, and the null player property. These axioms are sometimes taken to characterize fairness, although this view is not universally shared and does not align with the perspective adopted in this paper. Our view, shared by other scholars (see, e.g., [1]), is that a minimum requirement for a CAM to be considered fair is to provide an allocation which is in the core of the corresponding cooperative game (Property 7 below). Since the Shapley value does not fulfill this requirement, we would not qualify it as fair.

The individual allocated cost for member a under the Shapley CAM would be:

$$c_a^{Sh} = c_a^0 - v_a^{Sh} \quad (9)$$

- Distribution of savings using the nucleolus

Like the Shapley CAM, the nucleolus CAM departs from the definition of the coalitional game with transferrable payoff (A, v) introduced at the beginning of this Section 2.3.2. The nucleolus of this game provides a way of distributing the total savings $v(A) = (p_b - p_s)E^{Tr}$ among the members of the community A .

The nucleolus was introduced by Schmeidler [23] as a single-point solution concept for cooperative games. Intuitively, it is the (unique) allocation that minimizes the maximum dissatisfaction across all coalitions. Formally, the nucleolus is the unique distribution that lexicographically minimizes the vector of nonincreasingly ordered excesses of all possible coalitions over the set of possible distributions. The excess of a coalition at a certain distribution is the difference between the value that the coalition could obtain by deviating (i.e., forming their own EC, in our context) and the value given to the coalition under the distribution, so it can be understood as the dissatisfaction of the coalition with the current distribution (Note that the term excess is sometimes defined as the opposite. See, e.g., [24,25]).

The nucleolus has been proposed in the literature as a fair division scheme [26]. Its main limitation, however, lies in its computational complexity. In the general case, finding the nucleolus is highly demanding (see e.g., [7,24]); therefore, unless a more efficient algorithm is discovered for this EC particular setting, its practical applicability will remain limited (see also [1,3]).

2.3.3. Family of Price-Based Allocation Methods

The family of price-based allocation methods [9] distributes the total net cost by creating an internal energy market where net producers (i.e., supply) sell their excess to net consumers (i.e., demand) at a certain transfer price $p_{tr} \in [p_s, p_b]$. If demand does not match supply, the greater of the two is prorated. Once the market is cleared, any energy requirement or excess that agents may still have is individually traded with the grid at the

corresponding price. Formally, the allocated cost for member $a \in A$ under any price-based CAM is given by:

$$c_a^{PB} = \begin{cases} c_a^0 - \frac{e_a^{NC}}{\sum_{i \in NC} e_i^{NC}} E^{Tr}(p_b - p_{tr}) & \text{If } a \in NC \\ c_a^0 & \text{If } a \in N0 \\ c_a^0 - \frac{e_a^{NP}}{\sum_{i \in NP} e_i^{NP}} E^{Tr}(p_{tr} - p_s) & \text{If } a \in NP \end{cases} \quad (10)$$

There are different ways of setting the transfer price at which energy trading occurs, leading to different price-based CAMs. The following sections explain two possibilities. Finally, note that the Bill-Sharing (BS) rule could be interpreted as a price-based CAM where the transfer price p_{tr} is set to zero, i.e., net producers are forced to give their excess energy to net consumers for free [9].

- **Average Price**

The Average Price (AP) allocation method is a price-based allocation method where the transfer price is set to the average of the buying and selling prices with the grid:

$$p_{tr}^{AP} = \frac{p_b + p_s}{2} \quad (11)$$

The Average Price CAM can also be interpreted as a savings-based CAM where savings are distributed proportional to energy transferred in the internal market [9].

- **Extreme Price**

In the Extreme Price (EP) allocation method, all the savings generated by internal sharing are given to the scarce side of the internal market, i.e., the net consumers if $E^c < E^s$ or the net producers if $E^s < E^c$. Within the chosen set, savings are distributed proportional to the energy shared by each agent in the set. This can be achieved by setting a transfer price equal to p_s if the EC produces more energy than it consumes ($E^s > E^c$), or to p_b if the EC produces less energy than it consumes ($E^s < E^c$). If it consumes exactly the energy it produces, the price is set to the average of the two prices. Formally,

$$p_{tr}^{EP} = \begin{cases} p_s & \text{if } E^s > E^c \\ \frac{p_b + p_s}{2} & \text{if } E^s = E^c \\ p_b & \text{if } E^s < E^c \end{cases} \quad (12)$$

2.4. Definition of Desirable Properties for Cost Allocation Methods

In this section we define several desirable properties for cost allocation methods. Recall that c_a is the final net cost assigned to member a .

- **P1.** Parity among equals. (“Equal Treatment of Equals”). Participants who have the same net consumption must bear the same cost. Formally, $\forall a1, a2 \in A$, if $e_{a1}^c - e_{a1}^s = e_{a2}^c - e_{a2}^s$, then $c_{a1} = c_{a2}$
- **P2.** Disparity among unequals. (“Unequal Treatment of Unequals”). Participants who have different net consumptions must bear different costs. Formally, $\forall a1, a2 \in A$, if $e_{a1}^c - e_{a1}^s \neq e_{a2}^c - e_{a2}^s$, then $c_{a1} \neq c_{a2}$
- **P3.** Beneficial individual participation. Every participant is better off inside the community than on its own. Formally, $\forall a \in A$: $c_a \leq c_a^0$, and there are some cases where the inequality is strict.

- **P4.** Environmental friendliness. Participants' allocated costs are monotonically increasing with respect to their individual consumption. Consider the allocated cost as a function of individual consumption, $c_a(e_a^c)$, keeping everything else constant. Formally, P4 reads: $\forall a \in A$, if $x > y$, then $c_a(x) > c_a(y)$.
The weak version of P4 reads: Participants' allocated costs are monotonically non-decreasing with respect to their individual consumption. Formally, the weak version of P4 reads: $\forall a \in A$, if $x \geq y$, then $c_a(x) \geq c_a(y)$.
- **P5.** Continuity. Every participant's allocated cost is a continuous function of every participant's individual consumption. Consider the allocated cost of an individual $a \in A$ as a function of every participant's individual consumption, $c_a(e_1^c, e_2^c, \dots, e_n^c)$, keeping everything else constant. P5 states that $\forall a \in A$, $c_a(e_1^c, e_2^c, \dots, e_n^c)$ is a continuous function.
- **P6.** Rank correlation between costs and net consumption. Across all participants, the higher a participant's net consumption $(e_a^c - e_a^g)$, the higher their allocated cost. Formally, $\forall a1, a2 \in A$, if $(e_{a1}^c - e_{a1}^g) > (e_{a2}^c - e_{a2}^g)$ then $c_{a1} > c_{a2}$.
The weak version of P6 reads: $\forall a1, a2 \in A$, if $(e_{a1}^c - e_{a1}^g) \geq (e_{a2}^c - e_{a2}^g)$ then $c_{a1} \geq c_{a2}$.
- **P7.** Beneficial group participation. No subset of participants can obtain a lower cost if they form their own community. To define this property formally, let $C(S)$ be the total net cost incurred by a community formed by the group of agents S (see Equation (4)). P7 states that $\forall S \subseteq A$, $\sum_{a \in S} c_a \leq C(S)$. In terms of cooperative game theory, this property implies that every allocation given by the method is in the core.

2.5. Procedure for Evaluating Property Compliance of Cost Allocation Methods

The compliance of each Cost Allocation Method (CAM) with the proposed properties is assessed solely through formal analytical reasoning, without the need for simulations or empirical data. Each CAM is examined through its mathematical formulation within a rigorous axiomatic framework, which allows us to determine precisely whether it satisfies each of the desirable properties defined in Section 2.4.

The procedure consists of deriving, for every CAM, the logical and algebraic conditions under which each property holds for the Energy Community archetype under consideration. This axiomatic approach ensures objectivity, transparency, and methodological consistency in assessing the theoretical properties of each method.

The results are presented in a compliance matrix in Section 3 below, which shows which properties are fulfilled by each CAM. This purely analytical methodology guarantees objectivity, consistency, and methodological clarity in the comparison of CAMs.

3. Results and Discussion

Table 1 reports, for each cost allocation method, whether it satisfies each of the seven desirable properties introduced in Section 2.4. Several insights emerge from this comparative analysis.

Table 1. Compliance of the seven desirable properties for each of the eight CAMs considered.

	P1	P2	P3	P4	P5	P6	P7
	Parity Among Equals	Disparity Among Unequals	Beneficial Individual Participation	Environmental Friendliness	Continuity	Rank Correlation Between Costs and Net Consumption	Beneficial Group Participation
Simple rules							
All Equal	Y	N	N	Y	Y	Y (Weak)	N
Bill Sharing	Y	N	N	Y (Weak)	Y	Y (Weak)	N
Savings based							
Pre-established shares β_a	N	N	Y	Y	Y	N	N
Proportional to coefficients α_a	N	N	Y	Y	Y	N	N
Shapley	Y	Y	Y	Y	Y	Y	N
Nucleolus	Y	?	Y	?	Y	?	Y
Price based							
Average price	Y	Y	Y	Y	Y	Y	N
Extreme price	Y	Y	Y	N	N	Y	Y

Legend: Y = YES; N = NO; Y (Weak) means compliance with the weak version of the property; ? = Unproved (we conjecture YES).

3.1. Simple Rules

The two simple rules, *All Equal* and *Bill Sharing*, fail to satisfy P3 (beneficial individual participation). Under these approaches, some agents—typically net producers—may be worse off inside the community than outside it, which undermines participation incentives and makes these rules unsuitable for most practical settings. In particular, under *Bill Sharing*, energy transferred within the community is effectively given away for free, meaning that net producers lose the income they would obtain by selling their excess generation to the grid [15].

3.2. Savings-Based CAMs

Within the family of savings-based methods, the *nucleolus* stands out as the only method that satisfies P7 (beneficial group participation), ensuring that no coalition has an incentive to leave the community. The nucleolus also satisfies P1, P3, P5, and P7 [23], but we have not been able to prove (or disprove) compliance with P2, P4, and P6. Based on extensive simulations and intuition, we conjecture that the nucleolus does, in fact, satisfy all seven properties.

The *Shapley* method dominates the two simpler savings-based CAMs (sharing according to β_a or α_a), since it satisfies all the properties those methods satisfy and additional ones. However, the *Shapley* CAM does not satisfy P7. That is, some groups of agents may achieve lower costs by forming a sub-community. This is a significant limitation—one that motivates our view (shared by others in the literature, e.g., [1]) that the Shapley value should not be automatically regarded as a “fair” allocation in this context. Furthermore, both the Shapley value and the nucleolus suffer from substantial computational complexity, and this can limit their usefulness in practical settings.

3.3. Price-Based CAMs

Price-based methods are operationally appealing because they are simple to implement and have very low computational demands. The *Average-Price* CAM is particularly noteworthy: it satisfies exactly the same set of properties as the *Shapley* CAM but is computationally trivial. Its main limitation—shared with the *Shapley* CAM—is that it fails to guarantee P7.

By contrast, the *Extreme-Price* CAM is the only method that satisfies P7 (core stability), aside from the nucleolus. In this method, all savings accrue to the scarce side of the internal market (net producers when total consumption exceeds generation, and net consumers in the opposite case). Although this allocation may seem severe, in many situations it is the only one that ensures compliance with P7. To understand this, let us define a critical participant as one whose departure alone reduces the community's total exchanged energy. A net consumer is critical if $e_a^{NC} > E^c - E^g$ and a net producer is critical if $e_a^{NP} > E^g - E^c$. Proposition 1 below shows that, unless there exist both critical net consumers and critical net producers simultaneously, the only allocation that ensures P7 is the one provided by the *Extreme-price* CAM. This implies that if no net consumer is critical or no net producer is critical, then the allocations provided by the *Extreme-price* CAM and by the *nucleolus* CAM are identical.

Proposition 1. *If no net consumer is critical or no net producer is critical, the only allocation that ensures P7 “beneficial group participation” is the one given by the Extreme-price CAM.*

The main drawback of the *Extreme-price* CAM is that it fails to satisfy P4 and P5. Importantly, this trade-off is inherent to the entire price-based family, as shown by the following two propositions:

Proposition 2. *No price-based CAM can satisfy both P4 and P7.*

Proposition 3. *No price-based CAM can satisfy both P5 and P7.*

Overall, for applications where computational feasibility is essential, one of the two price-based methods is preferable. The appropriate choice depends on whether priority is given to P7 (in which case *Extreme-Price* is the only option) or to properties such as environmental friendliness and continuity (in which case *Average-Price* is the better choice). The nucleolus remains an excellent theoretical benchmark because it satisfies both P5 and P7—something no price-based method can achieve—but its high computational burden (see e.g., [7,24]) limits its practical applicability unless more efficient EC-specific algorithms are developed.

4. Conclusions

This paper has examined several Cost Allocation Methods (CAMs) for Energy Communities through a unified axiomatic framework grounded in principles of fairness, environmental friendliness, and continuity. By evaluating eight commonly used methods against seven formally defined properties, the analysis reveals several structural trade-offs that characterize the design space.

Price-based CAMs stand out for their ease of implementation and excellent scalability. Among them, the *Average-Price* method is especially appealing: it satisfies the same axiomatic properties as the *Shapley* CAM while being computationally trivial, making it a suitable choice in practical contexts where transparency and simplicity are important. However, neither *Average-Price* nor *Shapley* satisfies Property P7—beneficial group participation—which means that some coalitions may have incentives to leave the community.

The *Extreme-Price* CAM is the only price-based method that ensures P7. This strong stability guarantee makes it valuable in settings where coalition formation and long-term participation are paramount. Yet this comes at a cost: the method necessarily violates P4 (environmental friendliness) and P5 (continuity). We proved that this trade-off is inherent

to the entire price-based family—any price-based method that ensures P7 must sacrifice P4 and P5.

Beyond price-based approaches, the *nucleolus* CAM emerges as a theoretically compelling benchmark. It satisfies several key properties—including P7—and we conjecture, based on formal reasoning and numerical evidence, that it satisfies all seven. The main barrier to its adoption is computational complexity: in general settings, computing the nucleolus is extremely demanding, and no efficient algorithm is currently available for the EC context.

Overall, the unified framework developed here clarifies the landscape of cost allocation in energy communities, helping designers and policymakers choose methods that best match their normative goals and computational constraints. The results presented in this paper provide a transparent basis for understanding which desirable principles can—or cannot—be simultaneously achieved.

Several promising avenues for future research arise from this study:

1. Formal characterization of the nucleolus.

A central open question is whether the nucleolus indeed satisfies all seven desirable properties. Proving or disproving this conjecture would significantly advance the theoretical foundations of EC cost allocation and clarify whether a “gold standard” CAM exists.

2. Design of computationally efficient algorithms for the nucleolus.

Given the nucleolus’s strong theoretical appeal, identifying algorithms that exploit the structural features of the coalitional game that characterizes energy sharing could dramatically expand its practical applicability.

3. Extending the axiomatic framework to more complex EC settings.

Future work could consider settings with dynamic pricing, storage, network constraints, household flexibility, or multi-period coupling. Extending the framework to such contexts would help evaluate the robustness of CAM properties under more realistic conditions.

4. Empirical validation using real-world EC data.

Applying the framework to real consumption-generation patterns from operational energy communities would help assess how theoretical properties translate into practical outcomes and user perceptions.

Pursuing these directions will strengthen the theoretical, computational, and practical foundations of cost allocation in energy communities and support the development of fair, stable, and transparent sharing arrangements.

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Abbreviations

The following abbreviations are used in this manuscript:

AE	All Equal (method)
AP	Average Price (method)
BS	Bill Sharing (method)
CAM	Cost Allocation Method
EP	Extreme Price (method)
EC	Energy Community
NC	Net Consumer
NP	Net Producer
PB	Price-Based (method)

Symbols

A	Energy community
a	Individual member of an energy community
p_b	Buying price from the grid
p_s	Selling price to the grid
p_{tr}	Internal transfer price
E^g	Total energy generated by the community
α_a	Fraction of E^g assigned to agent $a \in A$
e_a^g	Energy obtained by agent a from the generation unit, $e_a^g = \alpha_a E^g$
e_a^c	Energy consumed by agent a
E^c	Total energy consumed by the community
NC	Net Consumers
NP	Net Producers
e_a^{NC}	Overconsumption of agent a , $e_a^{NC} = [e_a^c - e_a^g]^+$
e_a^{NP}	Overproduction of agent a , $e_a^{NP} = [e_a^g - e_a^c]^+$
C^0	Total cost incurred by a community that does not allow internal energy sharing
C	Total cost incurred by a community that allows internal energy sharing
c_a^0	Cost for an individual prosumer a if internal energy sharing is not allowed
c_a	Final net cost assigned to member a
E^{Tr}	Energy shared internally, $E^{Tr} = \min(E^{NC}, E^{NP})$
$v(S)$	Total savings that subset S of community members would obtain by internal sharing if they decided to part ways and form their own community
β_a	Pre-established share for agent a

Appendix A. Proofs of Property Compliance of Cost Allocation Methods

Appendix A.1. Family of Simple Rules

- All Equal

Proposition A1. (AE1). *The All Equal (AE) allocation method satisfies P1: “Parity among equals”.*

Proof. Since the allocation outcome is exactly the same for everyone, the system guarantees equal treatment to everyone. $\square$

Proposition A2. (AE2). *The All Equal (AE) allocation method does not satisfy P2: “Disparity among unequals”.*

Proof. Since the allocation outcome is exactly the same for everyone, the system treats every participant equally. $\square$

Proposition A3. (AE3). *The All Equal (AE) allocation method does not satisfy P3: “Beneficial individual participation”.*

Proof. The cost for member a outside the community (Equation (2)) is $c_a^0 = p_b e_a^{NC} - p_s e_a^{NP}$. For a net producer, this is $c_{a \in NP}^0 = -p_s e_a^{NP} < 0$. By contrast, the cost for a net producer under AE is:

$$c_{a \in NP}^{AE} = \frac{C}{n} = \frac{p_b [E^c - E^g]^+ - p_s [E^g - E^c]^+}{n}$$

If the community as a whole consumes more than it generates ($E^c > E^g$), then $c_a^{AE} = \frac{p_b (E^c - E^g)}{n} > 0$. Thus, in that case, net producers would be better off outside the community. $\square$

Proposition A4. (AE4). *The All Equal (AE) allocation method satisfies P4: “Environmental friendliness”.*

Proof. The total net cost C (Equation (1)) is a strictly increasing function of individuals' consumption. If one individual increases their consumption in Δx , the increment in cost for the community is $p_b \Delta x$ if $E^c \geq E^g$, with slope $\frac{\Delta C}{\Delta e_a} = p_b > 0$; while if $E^g > E^c$ the local slope is $\lim_{\Delta x \rightarrow 0^+} \frac{\Delta C}{\Delta x} = p_s > 0$. Consequently, the total cost C is an increasing function of individual consumptions, and so is the individual cost, $c_a = \frac{C}{n}$. $\square$

Proposition A5. (AE5). *The All Equal (AE) allocation method satisfies P5: “Continuity”.*

Proof. The total net cost C (Equation (1)) is a composition of continuous functions of every participant's individual consumption. $\square$

Proposition A6. (AE6). *The All Equal (AE) allocation method satisfies the weak version of P6: “Rank correlation between costs and net consumption”.*

Proof. The total net cost of the community is shared equally regardless of individuals' net consumption. $\square$

Proposition A7. (AE7). *The All Equal (AE) allocation method does not satisfy P7: “Beneficial group participation”.*

Proof. Since the All Equal (AE) allocation method does not satisfy P3, it does not satisfy P7. $\square$

- Bill Sharing

Proposition A8. (BS1). *The Bill Sharing (BS) allocation method satisfies P1: “Parity among equals”.*

Proof. Trivial from the formula of c_a^{BS} . $\square$

Proposition A9. (BS2). *The Bill Sharing (BS) allocation method does not satisfy P2: “Disparity among unequals”.*

Proof. In an EC where $E^g \geq E^c$, the cost for NCs is $c_{a \in NC}^{BS} = 0$, so two NCs that satisfy the premise of P2 would have the same cost. $\square$

Proposition A10. (BS3). *The Bill Sharing (BS) allocation method does not satisfy P3: “Beneficial individual participation”.*

Proof. Consider an EC with one net producer with $e_a^{NP} = 1$ and one net consumer with $e_b^{NC} = 1$. In that case:

$$c_{a \in NP}^{BS} = 0 > -p_s e_a^{NP} = c_{a \in NP}^0$$

The net producer is worse off inside the community, since she is not obtaining any income for her overproduction, but she could sell it to the grid at $p_s > 0$. $\square$

Proposition A11. (BS4). *The Bill Sharing (BS) allocation method satisfies the weak version of P4: “Environmental friendliness”.*

Proof. We prove the statement only for Net Producers. The proof for other members is analogous. The partial derivative of Net Producer a ’s allocated cost $c_{a \in NP}^{BS}$ with respect to her consumption e_a^c is (The partial derivative is not continuous at $E^g = E^c$, but this does not represent a problem for our purposes since it is non-negative regardless of the direction you are approaching the discontinuity):

$$\frac{\partial c_{a \in NP}^{BS}}{\partial e_a^c} = \begin{cases} p_s \left(\frac{\sum_i e_i^{NP} - e_a^{NP}}{(\sum_i e_i^{NP})^2} (E^g - E^c) + \frac{e_a^{NP}}{\sum_i e_i^{NP}} \right) & \text{if } E^g > E^c \\ 0 & \text{if } E^g < E^c \end{cases}$$

It is easy to check that this partial derivative is non-negative over its domain. Also, note that if $E^g < E^c$, then $\frac{\partial c_{a \in NP}^{BS}}{\partial e_a^c} = 0$, i.e., Net Producers may marginally change their consumption—so they are still Net Producers—without changing their allocated cost. Therefore, the BS CAM satisfies the weak version of P4, but not P4. $\square$

Proposition A12. (BS5). *The Bill Sharing (BS) allocation method satisfies P5: “Continuity”.*

Proof. The allocated cost c_a^{BS} is a composition of continuous functions of every participant’s individual consumption. $\square$

Proposition A13. (BS6). *The Bill Sharing (BS) allocation satisfies the weak version of P6: “Rank correlation between costs and net consumption”.*

Proof. In an EC where $E^g \geq E^c$, the cost for NCs is $c_{a \in NC}^{BS} = 0$, regardless of their net consumption. Thus, two NCs with different net consumption would be allocated the same cost. This implies that the BS CAM does not satisfy the strict version of P6.

However, the BS CAM does satisfy the weak version of P6. Observe that: each NP (for whom $e_{a \in NP}^c - e_{a \in NP}^g < 0$) is assigned a non-positive cost, since $e_{a \in NP}^{NC} = 0$ and $e_{a \in NP}^{NP} > 0$; each N0 (for whom $e_{a \in N0}^c - e_{a \in N0}^g = 0$) is assigned a nil cost, since $e_{a \in N0}^{NC} = 0$ and $e_{a \in N0}^{NP} = 0$; and each NC (for whom $e_{a \in NC}^c - e_{a \in NC}^g > 0$) is assigned a non-negative cost, since $e_{a \in NC}^{NC} > 0$ and $e_{a \in NC}^{NP} = 0$.

Thus, we only have to prove the weak version within each of the three sets, i.e., NP, N0 and NC. For N0, all agents have the same (nil) cost. For NPs:

$$c_{a \in NP}^{BS} = \begin{cases} -\frac{e_a^{NP}}{\sum_i e_i^{NP}} p_s (E^g - E^c) = \left(e_a^c - e_a^g \right) \frac{p_s (E^g - E^c)}{\sum_i e_i^{NP}} & \text{If } E^g \geq E^c \\ 0 & \text{If } E^g < E^c \end{cases}$$

Note that the term $\frac{p_s(E^g - E^c)}{\sum_i e_i^{NP}}$ is non-negative if $E^g \geq E^c$ and the same for all agents. And finally, for NCs:

$$c_{a \in NC}^{BS} = \begin{cases} 0 & \text{If } E^g \geq E^c \\ \frac{e_a^{NC}}{\sum_i e_i^{NC}} p_b(E^c - E^g) = (e_a^c - e_a^g) \frac{p_b(E^c - E^g)}{\sum_i e_i^{NC}} & \text{If } E^g < E^c \end{cases}$$

Note that the term $\frac{p_b(E^c - E^g)}{\sum_i e_i^{NC}}$ is non-negative if $E^g < E^c$ and the same for all agents. $\square$

Proposition A14. (BS7). *The Bill Sharing (BS) allocation method does not satisfy P7: “Beneficial group participation”.*

Proof. Since the Bill Sharing (BS) allocation method does not satisfy P3, it does not satisfy P7. $\square$

Appendix A.2. Family of Savings-Based Allocation Methods

- Distribution of savings proportional to pre-established shares (β CAM)

Proposition A15. (B1). *The β CAM does not satisfy P1: “Parity among equals”.*

Proof. Consider two agents $a1, a2 \in A$ that satisfy the premise of P1, i.e., $e_{a1}^c - e_{a1}^g = e_{a2}^c - e_{a2}^g$. This implies that $c_{a1}^0 = c_{a2}^0$. Thus, if $\beta_{a1} \neq \beta_{a2}$ and $E^{Tr} > 0$, then $c_{a1}^\beta \neq c_{a2}^\beta$. $\square$

Proposition A16. (B2). *The β CAM does not satisfy P2: “Disparity among unequals”.*

Proof. We provide a counterexample. Consider an EC comprising three agents $a1, a2, a3$. Assume that $e_{a1}^g = e_{a2}^g = e_{a3}^g = 3$, $e_{a1}^c = 4$, $e_{a2}^c = 5$ and $e_{a3}^c = 0$. This implies that $e_{a1}^{NC} = 1$, $e_{a1}^{NP} = 0$, $e_{a2}^{NC} = 2$, $e_{a2}^{NP} = 0$, $e_{a3}^{NC} = 0$, $e_{a3}^{NP} = 3$, and $E^{Tr} = 3$. Assume also that $p_s = 1$ and $p_b = 2$. Thus,

$$c_{a1}^\beta = p_b e_{a1}^{NC} - p_s e_{a1}^{NP} - \beta_{a1}(p_b - p_s)E^{Tr} = 2 - 3\beta_{a1}$$

$$c_{a2}^\beta = p_b e_{a2}^{NC} - p_s e_{a2}^{NP} - \beta_{a2}(p_b - p_s)E^{Tr} = 4 - 3\beta_{a2}$$

Note that $a1, a2$ satisfy the premise of P2. However, if $\beta_{a1} = 1/9$ and $\beta_{a2} = 7/9$, then $c_{a1}^\beta = c_{a2}^\beta = 5/3$. $\square$

Proposition A17. (B3). *The β CAM satisfies P3: “Beneficial individual participation”.*

Proof. $c_a^\beta = c_a^0 - \beta_a(p_b - p_s)E^{Tr} \leq c_a^0$. And if $\beta_a > 0$ and $E^{Tr} > 0$, then the inequality is strict. $\square$

Proposition A18. (B4). *The β CAM satisfies P4: “Environmental friendliness”.*

Proof. The allocated cost under the β CAM is:

$$c_a^\beta = p_b e_a^{NC} - p_s e_a^{NP} - \beta_a(p_b - p_s) \min\left(\sum_{a \in A} e_a^{NC}, \sum_{a \in A} e_a^{NP}\right)$$

Thus, for $a \in NC$, we have $\Delta c_{a \in NC}^\beta \geq \Delta e_a^c(p_b - \beta_a(p_b - p_s)) \geq \Delta e_a^c p_s$.

Similarly, for $a \in NP$, we have that $\Delta c_{a \in NP}^\beta \geq \Delta e_a^c p_s$. $\square$

Proposition A19. (B5). *The β CAM satisfies P5: “Continuity”.*

Proof. Since the savings obtained by internal sharing is continuous in every individual’s consumption, the β CAM is also continuous. $\square$

Proposition A20. (B6). *The β CAM does not satisfy P6: “Rank correlation between costs and net consumption”.*

Proof. Consider the counterexample in the proof of Proposition A16 (B2), but with $\beta_{a1} = 1/9$ and $\beta_{a2} = 5/6$. In this case, $(e_{a2}^c - e_{a2}^s) = 2 > 1 = (e_{a1}^c - e_{a1}^s)$, but:

$$c_{a2}^\beta = 4 - 3\beta_{a2} = 1.5 < 5/3 = 2 - 3\beta_{a1} = c_{a1}^\beta$$

$\square$

Proposition A21. (B7). *The β CAM does not satisfy P7: “Beneficial group participation”.*

Proof. Consider the counterexample in the proof of Proposition A20 (B6). Since $\beta_{a1} = 1/9$ and $\beta_{a2} = 5/6$, we have that $\beta_{a3} = 1 - \beta_{a1} - \beta_{a2} = 1/18$, and $c_{a3}^\beta = -p_s e_{a3}^{NP} - \beta_{a3}(p_b - p_s)E^{Tr} = -3 - \frac{3}{18} = -19/6$. In this case, agents $a1$ and $a3$ would be better off forming their own community $S = \{a1, a3\}$ since

$$\sum_{a \in S} c_a = c_{a1}^\beta + c_{a3}^\beta = \frac{5}{3} - \frac{19}{6} = -\frac{3}{2} > C(S) = -2p_s = -2$$

$\square$

- Distribution of savings proportional to coefficients α_a (α CAM)

Proposition A22. (PA1). *The α CAM does not satisfy P1: “Parity among equals”.*

Proof. We provide a counterexample. Consider two agents $a1, a2 \in A$ that satisfy the premise of P1, i.e., $e_{a1}^c - e_{a1}^s = e_{a2}^c - e_{a2}^s$, but with $\alpha_{a1} \neq \alpha_{a2}$. This is always possible because e_a^c does not depend on α_a . In this situation, $c_{a1}^0 = c_{a2}^0$, and if $E^{Tr} > 0$, then $c_{a1}^\alpha \neq c_{a2}^\alpha$. $\square$

Proposition A23. (PA2). *The α CAM does not satisfy P2: “Disparity among unequals”.*

Proof. We provide a counterexample. Consider an EC comprising three agents $a1, a2, a3$. Assume that $\alpha_{a1} = \alpha_{a2} = \frac{1}{9}$, $\alpha_{a3} = \frac{7}{9}$. $E^s = 27$, so $e_{a1}^s = e_{a2}^s = 3$, and $e_{a3}^s = 21$. Assume $e_{a1}^c = 4$, $e_{a2}^c = 0$ and $e_{a3}^c = 23$. This implies that $e_{a1}^{NC} = 1$, $e_{a2}^{NP} = 3$, $e_{a3}^{NC} = 2$ and $E^{Tr} = 3$. Assume also that $p_s = 1$ and $p_b = 2$. Thus,

$$c_{a1}^\alpha = p_b e_{a1}^{NC} - p_s e_{a1}^{NP} - \alpha_{a1}(p_b - p_s)E^{Tr} = 2 - \frac{1}{9}3 = 5/3$$

$$c_{a3}^\alpha = p_b e_{a3}^{NC} - p_s e_{a3}^{NP} - \alpha_{a3}(p_b - p_s)E^{Tr} = 4 - \frac{7}{9}3 = 5/3$$

Note that $a1, a3$ satisfy the premise of P2 but $c_{a1}^\alpha = c_{a3}^\alpha = 5/3$.

The premise of P2 implies $e_{a1}^s = e_{a2}^s$. Given that $e_a^s = \alpha_a E^s$, we have that $\alpha_{a1} = \alpha_{a2}$. Given that $e_{a1}^c \neq e_{a2}^c$, we have $c_{a1}^0 \neq c_{a2}^0$, and therefore $c_{a1}^\alpha = c_{a1}^0 - \alpha_{a1}(p_b - p_s)E^{Tr} \neq c_{a2}^\alpha$. $\square$

Proposition A24. (PA3). *The α CAM satisfies P3: “Beneficial individual participation”.*

Proof. See proof of Proposition A17 (B3). $\square$

Proposition A25. (PA4). *The α CAM satisfies P4: “Environmental friendliness”.*

Proof. See proof of Proposition A18 (B4). $\square$

Proposition A26. (PA5). *The α CAM satisfies P5: “Continuity”.*

Proof. See proof of Proposition A19 (B5). $\square$

Proposition A27. (PA6). *The α CAM does not satisfy P6: “Rank correlation between costs and net consumption”.*

Proof. We provide a counterexample. Consider an EC comprising three agents $a1$, $a2$, $a3$. Assume that $E^g = 100$, $\alpha_{a1} = \alpha_{a2} = 0.01$ and $\alpha_{a3} = 0.98$. Thus, $e_{a1}^g = e_{a2}^g = 1$ and $e_{a3}^g = 98$. Assume $e_{a1}^c = 2$, $e_{a2}^c = 0$ and $e_{a3}^c = 98$. This implies that $e_{a1}^{NC} = 1$, $e_{a2}^{NP} = 1$, $e_{a3}^{NC} = 0$, $e_{a3}^{NP} = 0$, and $E^{Tr} = 1$. Assume also that $p_s = 1$ and $p_b = 3$. Thus,

$$(e_{a3}^c - e_{a3}^g) = 0 > -1 = (e_{a2}^c - e_{a2}^g)$$

However, $c_{a3}^\alpha < c_{a2}^\alpha$, as shown below:

$$c_{a3}^\alpha = p_b e_{a3}^{NC} - p_s e_{a3}^{NP} - \alpha_{a3}(p_b - p_s)E^{Tr} = -0.98 \times 2 \times 1 = -1.96$$

$$c_{a2}^\alpha = p_b e_{a2}^{NC} - p_s e_{a2}^{NP} - \alpha_{a2}(p_b - p_s)E^{Tr} = -1 \times 1 - 0.01 \cdot 2 \times 1 = -1.02$$

$\square$

Proposition A28. (PA7). *The α CAM does not satisfy P7: “Beneficial group participation”.*

Proof. Consider the same counterexample as in the proof of Proposition A27 (PA6). Note that

$$c_{a1}^\alpha = p_b e_{a1}^{NC} - p_s e_{a1}^{NP} - \alpha_{a1}(p_b - p_s)E^{Tr} = 3 \times 1 - 0.01 \times 2 \times 1 = 2.98$$

In this case, agents $a1$ and $a2$ would be better off forming their own community, since:

$$c_{a1}^\alpha + c_{a2}^\alpha = 2.98 - 1.02 = 1.96 > 0 = C(\{a1, a2\})$$

$\square$

- Distribution of savings using the Shapley value

Proposition A29. (S1). *The Shapley CAM satisfies P1: “Parity among equals”.*

Proof. The Shapley value satisfies the property “symmetry”, also known as “equal treatment of equals”, which reads: if $i \in A$ and $j \in A$ are equivalent in the sense that $v(S \cup \{i\}) = v(S \cup \{j\})$ for every subset $S \subset A$ which contains neither i nor j , then $v_i^{Sh} = v_j^{Sh}$.

If two agents satisfy the premise of P1, their marginal contributions to every (sub-) community will be the same, so their cost under the Shapley CAM will be the same.

$$c_a^{Sh} = e_a^{NC} p_b - e_a^{NP} p_s - v_a^{Sh}$$

□

Proposition A30. (S2). *The Shapley CAM satisfies P2: “Disparity among unequals”.*

Proof. Importantly, note that, given that $p_s > 0$, we have that $0 < v_a^{Sh} < e_a^{NC} p_b$ for all $a \in NC$. In words, this means that the share of the surplus that a NC is going to get under the Shapley CAM is necessarily lower than the total cost of the overconsumption if the NC were to buy her overconsumption from the grid. The reason is that the marginal contribution of $a \in NC$ at any coalition, which is the increase in surplus for that coalition caused by the introduction of $a \in NC$, is bounded below by 0 and above by $e_a^{NC}(p_b - p_s)$. Since v_a^{Sh} is an average of the marginal contributions, and $p_s > 0$, we have

$$0 \leq v_a^{Sh} \leq e_a^{NC}(p_b - p_s) < e_a^{NC} p_b$$

Now, consider two agents $a1$ and $a2$ such that satisfy the premise of P2, i.e., $e_{a1}^c - e_{a1}^s \neq e_{a2}^c - e_{a2}^s$. Recall that each agent can only belong to one of the three sets: NP , NC or $N0$. If the two agents belong to a different set, then their corresponding costs will be different, since

$$\begin{aligned} c_{a \in NC}^{Sh} &= c_{a \in NC}^0 - v_a^{Sh} = p_b e_a^{NC} - v_a^{Sh} > 0 \\ c_{a \in N0}^{Sh} &= c_{a \in N0}^0 - v_a^{Sh} = 0 \\ c_{a \in NP}^{Sh} &= c_{a \in NP}^0 - v_a^{Sh} = -p_s e_a^{NP} - v_a^{Sh} < 0 \end{aligned}$$

So, in that case, the proposition would be true.

Agents $a1$ and $a2$ cannot belong both to $N0$, since $e_{a1}^c - e_{a1}^s \neq e_{a2}^c - e_{a2}^s$.

Thus, it only remains to consider the cases where the two agents belong both to NP or both to NC . Let us first suppose that they both belong to NP .

Without loss of generality, assume that $e_{a1}^{NP} > e_{a2}^{NP}$. The marginal contribution of an agent to any given coalition is the change it causes in $\min(\sum e^{NP}, \sum e^{NC})(p_b - p_s)$. Considering the marginal contribution of $a1$ and $a2$ to corresponding coalitions, under a one-to-one correspondence between the sets of coalitions over which v_{a1}^{Sh} and v_{a2}^{Sh} are computed (Note that the computation of v_{a1}^{Sh} is performed over the set of coalitions formed by agents in $S_{a1} = A \setminus \{a1\}$, while the computation of v_{a2}^{Sh} is performed over the set of coalitions formed by agents in $S_{a2} = A \setminus \{a2\}$. Let $S_{a1,a2} = A \setminus \{a1, a2\}$ and note that $S_{a1,a2} \subset S_{a1}$ and $S_{a1,a2} \subset S_{a2}$. Under the one-to-one correspondence between S_{a1} and S_{a2} , every coalition in $S_{a1,a2}$ maps to itself; and a coalition $\rho_1 \subseteq S_{a1}$ such that $a2 \in \rho_1$ maps to the coalition $\rho_2 \in S_{a2}$ where agent $a2$ in ρ_1 has been replaced with $a1$), and noting that v_a^{Sh} is an average of marginal contributions, we find

$$0 \leq v_{a1}^{Sh} - v_{a2}^{Sh} \leq (e_{a1}^{NP} - e_{a2}^{NP})(p_b - p_s)$$

From $0 \leq v_{a1}^{Sh} - v_{a2}^{Sh}$, we have:

$$c_{a1 \in NP}^{Sh} - c_{a2 \in NP}^{Sh} = -p_s (e_{a1}^{NP} - e_{a2}^{NP}) - (v_{a1}^{Sh} - v_{a2}^{Sh}) < 0$$

Thus, the costs for agents $a1$ and $a2$ are indeed different.

For two net consumers with $e_{a1}^{NC} > e_{a2}^{NC}$ we obtain $0 \leq v_{a1}^{Sh} - v_{a2}^{Sh} \leq (e_{a1}^{NC} - e_{a2}^{NC})(p_b - p_s)$

From $(v_{a1}^{Sh} - v_{a2}^{Sh}) \leq (e_{a1}^{NC} - e_{a2}^{NC})(p_b - p_s)$, we have

$$c_{a1 \in NC}^{Sh} - c_{a2 \in NC}^{Sh} = p_b(e_{a1}^{NC} - e_{a2}^{NC}) - (v_{a1}^{Sh} - v_{a2}^{Sh}) \geq p_s(e_{a1}^{NC} - e_{a2}^{NC}) > 0$$

□

Proposition A31. (S3). *The Shapley CAM satisfies P3: “Beneficial individual participation”.*

Proof. Since savings cannot be negative, $v_a^{Sh} \geq 0$, so it is always the case that $c_a^{Sh} = c_a^0 - v_a^{Sh} \leq c_a^0$ for all $a \in A$. Furthermore, if there is at least one net consumer and one net producer, then $v_a^{Sh} > 0$ for all net consumers and producers, and therefore $c_a^{Sh} < c_a^0$. □

Proposition A32. (S4). *The Shapley CAM satisfies P4: “Environmental friendliness”.*

Proof. We show that an increase in individual consumption necessarily leads to an increase in allocated cost. For NPs, we have

$$c_{a \in NP}^{Sh} = c_{a \in NP}^0 - v_a^{Sh} = -(p_s e_a^{NP} + v_a^{Sh})$$

Following the same reasoning as in the proof of Proposition A30 (S2), we find that, if there is an increment Δe_a^c (i.e., a reduction in e_a^{NP}), keeping the participant as a net producer, then v_a^{Sh} does not increase while $p_s e_a^{NP}$ decreases, so the cost $-(p_s e_a^{NP} + v_a^{Sh})$ increases.

For NCs and agents in $N0$, we have

$$c_a^{Sh} = p_b e_a^{NC} - v_a^{Sh}$$

For $\Delta e_a^c > 0$, following the same line of reasoning as in the proof of Proposition A30 (S2), we have $\frac{\Delta v_a^{Sh}}{\Delta e_a^c} \leq p_b - p_s$, so

$$\frac{\Delta c_a^{Sh}}{\Delta e_a^c} = p_b - \frac{\Delta v_a^{Sh}}{\Delta e_a^c} \geq p_b - (p_b - p_s) = p_s > 0$$

□

Proposition A33. (S5). *The Shapley CAM satisfies P5: “Continuity”.*

Proof. The savings obtained by internal sharing are continuous in every individual’s consumption. Thus, the Shapley value v_a^{Sh} , which is the average increase in surplus caused by the introduction of an agent in a market, is also continuous in every individual’s consumption. Thus, c_a^S is continuous. □

Proposition A34. (S6). *The Shapley CAM satisfies P6: “Rank correlation between costs and net consumption”.*

Proof. Under the Shapley CAM, the cost for any net consumer is greater than the cost for any agent in $N0$, which in turn is greater than the cost for any Net Producer (see proof of Proposition S.2). The proof for agents in $N0$ is trivial, since they all have net consumption equal to zero and they are all allocated the same cost. Thus, it only remains to consider the cases where the two agents belong both to NP or both to NC .

The proof of the statement for Net Producers is included in the proof of Proposition A30 (S2), where it is shown that $\forall a1, a2 \in NP, e_{a1}^{NP} > e_{a2}^{NP} \implies c_{a1}^{Sh} < c_{a2}^{Sh}$. The proof for Net Consumers is analogous. $\square$

Proposition A35. (S7). *The Shapley CAM does not satisfy P7: “Beneficial group participation”.*

Proof. We can use as a counterexample any “glove game” [27,28] with different numbers of left-hand owners and right-hand owners. In a glove game, each agent owns a single glove (either left- or right-hand) and the worth of a coalition is the number of complete pairs it contains. In these games, the scarce side captures all the value in the core.

The analogy in our setting would be a market where each NC demands one unit of energy and each NP supplies one unit. If the numbers of NCs and NPs are different (and both positive), the core collapses to the allocation in which the entire surplus is distributed uniformly among the agents on the scarce side of the market. By contrast, the Shapley value lies outside the core, since every agent on the abundant side still makes a positive marginal contribution to certain coalitions. $\square$

Appendix A.3. Family of Price-Based Allocation Methods

Proposition A36. (PB1). *Price-based (PB) allocation methods satisfy P1: “equal treatment among equals”.*

Proof. We prove the statement only for Net Consumers. The proof for Net Producers is analogous. The allocated cost for Net Consumers is:

$$c_{a \in NC}^{PB} = p_b e_a^{NC} - \frac{e_a^{NC}}{\sum_{i \in NC} e_i^{NC}} E^{Tr}(p_b - p_{tr})$$

It is clear that $\forall a1, a2 \in NC$, if $e_{a1}^{NC} = e_{a2}^{NC}$, then $c_{a1}^{PB} = c_{a2}^{PB}$. $\square$

Proposition A37. (PB2). *Price-based (PB) allocation methods with $p_{tr} \in [p_s, p_b]$ satisfy P2: “Disparity among unequals” and P6: “Rank correlation between costs and net consumption”.*

Proof. Consider two agents $a1$ and $a2$ that satisfy the premise of P2. Recall that each agent can only belong to one of the three sets: NP, NC or N0. If the two agents belong to a different set (and bearing in mind that $p_{tr} \in [p_s, p_b]$), then their corresponding costs will be different, since

$$\begin{aligned} c_{a \in NC}^{PB} &= p_b e_a^{NC} - \frac{e_a^{NC}}{\sum_{i \in NC} e_i^{NC}} E^{Tr}(p_b - p_{tr}) > 0 \\ c_{a \in N0}^{PB} &= 0 \\ c_{a \in NP}^{PB} &= -p_s e_a^{NP} - \frac{e_a^{NP}}{\sum_{i \in NP} e_i^{NP}} E^{Tr}(p_{tr} - p_s) < 0 \end{aligned}$$

So, in that case, the proposition would be true.

Agents $a1$ and $a2$ cannot belong both to N0, since $e_{a1}^c - e_{a1}^g \neq e_{a2}^c - e_{a2}^g$.

Thus, it only remains to consider the cases where the two agents belong both to NP or both to NC. The cost for NCs is positive and proportional to e_a^{NC} , so $\forall a1, a2 \in NC$, if $e_{a1}^{NC} > e_{a2}^{NC}$ (i.e., $(e_{a1}^c - e_{a1}^g) > (e_{a2}^c - e_{a2}^g)$) then $c_{a1}^{PB} > c_{a2}^{PB}$. Similarly, the cost for NPs is negative and its absolute value is proportional to e_a^{NP} , so $\forall a1, a2 \in NP$, if $e_{a1}^{NP} > e_{a2}^{NP}$ (i.e., $(e_{a1}^c - e_{a1}^g) < (e_{a2}^c - e_{a2}^g)$) then $c_{a1}^{PB} < c_{a2}^{PB}$. $\square$

Proposition A38. (PB3). Price-based (PB) allocation methods with $p_{tr} \in [p_s, p_b]$ satisfy P3: “Beneficial individual participation”.

Proof. For Net Producers we have: $c_{a \in NP}^0 - c_{a \in NP}^{PB} = \frac{e_a^{NP}}{\sum_{i \in NP} e_i^{NP}} E^{Tr} (p_{tr} - p_s) \geq 0$. This inequality is strict whenever there is trading in the internal market ($E^{Tr} > 0$) and $p_{tr} > p_s$. Analogously, for Net Consumers: $c_{a \in NC}^0 - c_{a \in NC}^{PB} = \frac{e_a^{NC}}{\sum_{i \in NC} e_i^{NC}} E^{Tr} (p_b - p_{tr}) \geq 0$. And the inequality is strict whenever there is trading in the internal market ($E^{Tr} > 0$) and $p_b > p_{tr}$. $\square$

Proposition A39. (PB4). Price-based (PB) allocation methods where p_{tr} is a continuous function of individuals’ consumption satisfy P5: “Continuity”.

Proof. Given that p_{tr} is a continuous function of every participants’ individual cost (by assumption), the formula for c_a^{PB} is a composition of continuous functions of every participants’ individual cost. $\square$

- Average Price

Proposition A40. (AP1). The Average Price (AP) CAM satisfies P1: “equal treatment among equals”.

Proof. See Proposition A36 (PB1). $\square$

Proposition A41. (AP2). The Average Price (AP) allocation method satisfies P2: “Disparity among unequals”.

Proof. Since $p_{tr}^{AP} \in [p_s, p_b]$, we can use Proposition A37 (PB2). $\square$

Proposition A42. (AP3). The Average Price (AP) allocation method satisfies P3: “Beneficial individual participation”.

Proof. Since $p_{tr}^{AP} \in [p_s, p_b]$, we can use Proposition A38 (PB3). $\square$

Proposition A43. (AP4). The Average Price (AP) allocation method satisfies P4: “Environmental friendliness”.

Proof. For a net consumer $a \in NC$, starting from an initial situation in which the traded energy is E_1^{Tr} and the total overconsumption is $\sum_i e_i^{NC} = E_1^{NC} > 0$ we have

$$c_{a \in NC}^{PB} = e_a^{NC} \left(p_b - \frac{E_1^{Tr}}{E_1^{NC}} \left(\frac{p_b - p_s}{2} \right) \right)$$

If net consumer $a \in NC$ increases her consumption in Δe_a^c , the new cost is

$$(e_a^{NC} + \Delta e_a^c) \left(p_b - \frac{E_1^{Tr}}{E_1^{NC} + \Delta e_a^c} \left(\frac{p_b - p_s}{2} \right) \right) \geq (e_a^{NC} + \Delta e_a^c) \left(p_b - \frac{E_1^{Tr} + \Delta e_a^c}{E_1^{NC}} \left(\frac{p_b - p_s}{2} \right) \right)$$

Consequently,

$$\begin{aligned} \Delta c_{a \in NC}^{PB} &\geq \Delta e_a^c \left(p_b - \left(\frac{e_a^{NC}}{E_1^{NC}} + \frac{E_1^{Tr}}{E_1^{NC}} + \frac{\Delta e_a^c}{E_1^{NC}} \right) \left(\frac{p_b - p_s}{2} \right) \right) \\ &\geq \Delta e_a^c \left(p_b - \left(2 + \frac{\Delta e_a^c}{E_1^{NC}} \right) \left(\frac{p_b - p_s}{2} \right) \right) \end{aligned}$$

And, finally,

$$\lim_{\Delta e_a^c \rightarrow 0^+} \frac{\Delta c_{a \in NC}^{PB}}{\Delta e_a^c} \geq \lim_{\Delta e_a^c \rightarrow 0^+} \left(p_b - \left(2 + \frac{\Delta e_a^c}{E_1^{NC}} \right) \left(\frac{p_b - p_s}{2} \right) \right) = p_s$$

Analogously, the same lower bound (p_s) for $\lim_{\Delta e_a^c \rightarrow 0^+} \frac{\Delta c_{a \in NP}^{PB}}{\Delta e_a^c}$ can be shown to apply to net producers and participants in $N0$. $\square$

Proposition A44. (AP5). *The Average Price (AP) allocation method satisfies P5: “Continuity”.*

Proof. Price p_{tr}^{AP} is independent of individuals’ consumption, so we can use Proposition A39 (PB4) to show that the AP allocation method satisfies P5: “Continuity”. $\square$

Proposition A45. (AP6). *The Average Price (AP) allocation method satisfies P6: “Rank correlation between costs and net consumption”.*

Proof. Since $p_{tr}^{AP} \in [p_s, p_b]$, we can use Proposition A37 (PB2). $\square$

Proposition A46. (AP7). *The Average Price (AP) allocation method does not satisfy P7: “Beneficial group participation”.*

Proof. We can use the fact that the AP CAM satisfies P4 and P5, together with Proposition 2 or Proposition 3. Nonetheless, here we provide a counterexample to provide the intuition why the AP CAM does not fulfill P7.

Consider the following “glove market” [27,28] with three agents $a1$, $a2$ and $a3$, with the following data: $e_{a1}^s = e_{a2}^s = e_{a3}^s = 3$; $e_{a1}^c = 5$; $e_{a2}^c = e_{a3}^c = 1$; $p_b = 30$; $p_s = 10$. With these conditions, $e_{a1}^{NC} = 2$ and $e_{a2}^{NP} = e_{a3}^{NP} = 2$. Agent $a1$ would obtain 1 unit of energy from $a2$ and 1 unit of energy from $a3$ at $p_{tr}^{AP} = \frac{p_b + p_s}{2} = 20$. The net costs for the members would be:

$$\begin{aligned} c_{a1 \in NC}^{AP} &= e_{a1}^{NC} p_b - E^{Tr} \left(p_b - p_{tr}^{AP} \right) = 2 \times 30 - 2 \times 10 = 40 \\ c_{a2 \in NP}^{AP} &= -e_{a2}^{NP} p_s - \frac{e_{a2}^{NP}}{e_{a2}^{NP} + e_{a3}^{NP}} E^{Tr} \left(p_{tr}^{AP} - p_s \right) = -2 \times 10 - \frac{2}{4} 2 \times 10 = -30 \\ &= c_{a3 \in NP}^{AP} \end{aligned}$$

Thus, the subgroup formed by $a1$ and $a2$ bears a total net cost of $40 - 30 = 10$. However, if they formed their own community (leaving $a3$ behind), the total net cost of that coalition would be $C(\{a1, a2\}) = 0$. $\square$

- Extreme Price

Proposition A47. (EP1). *The Extreme Price (EP) allocation method satisfies P1: “equal treatment among equals”.*

Proof. See Proposition A36 (PB1). $\square$

Proposition A48. (EP2). *The Extreme Price (EP) allocation method satisfies P2: “Disparity among unequals”.*

Proof. Since $p_{tr}^{EP} \in [p_s, p_b]$, we can use Proposition A37 (PB2). $\square$

Proposition A49. (EP3). *The Extreme Price (EP) allocation method satisfies P3: “Beneficial individual participation”.*

Proof. Since $p_{tr}^{EP} \in [p_s, p_b]$, we can use Proposition A38 (PB3). $\square$

Proposition A50. (EP4). *The Extreme Price (EP) allocation method does not satisfy P4: “Environmental friendliness”.*

Proof. We can use the fact that the EP CAM satisfies P7 (Proposition A53 (EP7)), together with Proposition 2. Nonetheless, here we provide a counterexample to provide the intuition why the EP CAM does not fulfill P4.

Under the EP CAM, an increase in an individual’s consumption may lead to an increase in p_{tr} from p_s to p_b . This benefits net producers. In fact, under the EP CAM, it may be the case that an individual’s increase in energy consumption (keeping everything else constant) may lead to a reduction in her individual allocated cost.

As an example, consider a market with two agents $a1$ and $a2$, with the following data: $e_{a1}^g = e_{a2}^g = 2$; $e_{a1}^c = 3$; $e_{a2}^c = 0.9$; $p_b = 100$; $p_s = 10$. With these conditions, $e_{a1}^{NC} = 1$ and $e_{a2}^{NP} = 1.1$. Agent $a1$ would obtain 1 unit of energy from agent $a2$ at $p_{tr}^{EP} = p_s = 10$. The net cost for $a2$ would be:

$$c_{a2 \in NP}^{EP} = -e_{a2}^{NP} p_s - E^{Tr} (p_{tr}^{EP} - p_s) = -1.1 \times 10 - 1 \times 0 = -11$$

If now $a2$ increases consumption to $e_{a2}^{c,*} = 1.1$, we have $e_{a1}^{NC,*} = 1$ and $e_{a2}^{NP,*} = 0.9$. Agent $a1$ would obtain 0.9 units of energy from agent $a2$ at $p_{tr}^{EP,*} = p_b = 100$. The new net cost for $a2$ would be:

$$c_{a2 \in NP}^{EP,*} = -e_{a2}^{NP,*} p_s - E^{Tr,*} (p_{tr}^{EP,*} - p_s) = -0.9 \times 10 - 0.9 \times 90 = -90$$

Thus, $a2$ increased her consumption and her individual cost greatly decreased. $\square$

Proposition A51. (EP5). *The Extreme Price (EP) allocation method does not satisfy P5 “Continuity”.*

Proof. We provide a counterexample. Consider a market with two agents $a1$ and $a2$, with the following data: $e_{a1}^g = e_{a2}^g = 2$; $e_{a1}^c = 3$; $e_{a2}^c = 1$; $p_b = 100$; $p_s = 10$. With these conditions, $e_{a1}^{NC} = 1$ and $e_{a2}^{NP} = 1$. Agent $a1$ would obtain 1 unit of energy from agent $a2$ at $p_{tr}^{EP} = \frac{p_b + p_s}{2} = 55$. The net cost for $a2$ would be:

$$c_{a2 \in NP}^{EP} = -e_{a2}^{NP} p_s - \min(e_{a1}^{NC}, e_{a2}^{NP}) (p_{tr}^{EP} - p_s) = -p_{tr}^{EP} = -55$$

However, if the consumption e_{a2}^c increases, then $E^g < E^c$ and we have $p_{tr} = p_b = 100$, while if e_{a2}^c decreases then $E^g > E^c$ and $p_{tr} = p_s = 10$. Consequently,

$$\lim_{e_{a2}^c \rightarrow 1^+} c_{a2 \in NP}^{EP} = - \lim_{e_{a2}^c \rightarrow 1^+} (e_{a2}^{NP} p_s + e_{a1}^{NC} (p_{tr}^{EP} - p_s)) = - \lim_{e_{a2}^c \rightarrow 1^+} e_{a1}^{NC} p_b = -p_b = -100$$

$$\lim_{e_{a2}^c \rightarrow 1^-} c_{a2 \in NP}^{EP} = - \lim_{e_{a2}^c \rightarrow 1^-} e_{a2}^{NP} p_{tr}^{EP} = - \lim_{e_{a2}^c \rightarrow 1^-} e_{a2}^{NP} p_s = -p_s = -10$$

Note that this is also a proof that P4 “Environmental friendliness” is not satisfied, as $c_{a2 \in NP}^{EP}$ decreases from -55 to (approximately) -100 if $a2$ increases its consumption slightly. $\square$

Proposition A52. (EP6). *The Extreme Price (EP) allocation method satisfies P6: “Rank correlation between costs and net consumption”.*

Proof. Since $p_{tr}^{EP} \in [p_s, p_b]$, we can use Proposition A37 (PB2). $\square$

Proposition A53. (EP7). *The Extreme Price (EP) allocation method satisfies P6 “Beneficial group participation”.*

Proof. The savings obtained by a coalition $S \subseteq A$ that forms its own community equal $v(S) = (p_b - p_s) \min(\sum_{a \in S} e_a^{NC}, \sum_{a \in S} e_a^{NP})$. We must compare these savings with the sum of the individual savings obtained by the agents in S when they are part of the grand community A and savings are distributed according to the EP CAM. The individual saving for agent $a \in A$ under the EP CAM is denoted $v_a^{EP} = c_a^0 - c_a^{EP}$. Thus, our goal is to show that $\sum_{a \in S} v_a^{EP} \geq v(S) \forall S \subseteq A$.

First, note that $E^{NP} - E^{NC} = \sum_{a \in NP} [e_a^g - e_a^c]^+ - \sum_{a \in NC} [e_a^c - e_a^g]^+ = \sum_{a \in NP} (e_a^g - e_a^c) - \sum_{a \in NC} (e_a^c - e_a^g) = \sum_{a \in A} e_a^g - \sum_{a \in A} e_a^c = E^g - E^c$. This implies that the savings obtained by agent $a \in A$ under the EP CAM, $v_a^{EP} = c_a^0 - c_a^{EP}$, can be written as:

$$v_a^{EP} = \begin{cases} e_a^{NC}(p_b - p_s) & \text{if } E^{NP} > E^{NC} \\ (e_a^{NC} + e_a^{NP}) \left(\frac{p_b - p_s}{2} \right) & \text{if } E^{NP} = E^{NC} \\ e_a^{NP}(p_b - p_s) & \text{if } E^{NP} < E^{NC} \end{cases}$$

Thus, $\forall S \subseteq A$,

$$\sum_{a \in S} v_a^{EP} = \begin{cases} (p_b - p_s) \sum_{a \in S} e_a^{NC} & \text{if } E^{NP} > E^{NC} \\ \left(\frac{p_b - p_s}{2} \right) \sum_{a \in S} (e_a^{NC} + e_a^{NP}) & \text{if } E^{NP} = E^{NC} \\ (p_b - p_s) \sum_{a \in S} e_a^{NP} & \text{if } E^{NP} < E^{NC} \end{cases}$$

It is then clear that $\sum_{a \in S} v_a^{EP} \geq v(S) = (p_b - p_s) \min(\sum_{a \in S} e_a^{NC}, \sum_{a \in S} e_a^{NP}) \forall S \subseteq A$. $\square$

Appendix B. Proofs of Propositions in the Body of the Paper

Lemma A1. *Under any CAM that satisfies P7, if $a \in A$ is not critical, then $c_a = c_a^0$.*

Proof. We prove the statement assuming that the non-critical agent is a net consumer. The proof for non-critical net producers is analogous.

Suppose net consumer $a \in NC$ is not critical in community A , i.e., $e_a^{NC} \leq E_A^c - E_A^g = E_A^{NC} - E_A^{NP}$ (see proof of Proposition A53 (EP7)). Let E_A^{Tr} be the amount of energy traded in community A . Since $a \in NC$, $e_a^{NC} > 0$, so it must be the case that $E_A^{NC} > E_A^{NP}$, which implies $E_A^{Tr} = \sum_{i \in A} e_i^{NP} \leq \sum_{i \in A} e_i^{NC} - e_a^{NC} \implies \sum_{i \in A \setminus \{a\}} e_i^{NP} \leq \sum_{i \in A \setminus \{a\}} e_i^{NC} \implies E_A^{Tr} = E_{A \setminus \{a\}}^{Tr}$. In words, the amount of energy traded in community A does not change if a alone departs.

The individual savings assigned by the CAM to agent a in community A is denoted $v_a = c_a^0 - c_a$. Fulfilling P7 means that $\sum_{i \in S} v_i \geq E_S^{Tr}(p_b - p_s) \forall S \subseteq A$. Applying P7 to the sets formed by each individual $i \in A$ alone, we have that $v_i \geq 0$.

Applying P7 to $A \setminus \{a\}$, we have

$$\sum_{i \in A \setminus \{a\}} v_i \geq E_{A \setminus \{a\}}^{Tr}(p_b - p_s) = E_A^{Tr}(p_b - p_s) = \sum_{i \in A} v_i$$

This implies that $v_a = 0$, so $c_a = c_a^0$. $\square$

Lemma A2. *Under any price-based CAM that satisfies P7:*

- *If there is some non-critical net consumer d , i.e., $e_d^{NC} \leq E^c - E^g$, then $v_{a \in NC} = 0$, $v_{a \in NP} = e_a^{NP}(p_b - p_s)$ and $c_{a \in A} = (e_a^c - e_a^g)p_b$. Furthermore, if $E^{Tr} > 0$, then $p_{tr} = p_b$.*

- If there is some non-critical net producer h , i.e., $e_h^{NP} \leq E^g - E^c$, then $v_{a \in NP} = 0$, $v_{a \in NC} = e_a^{NC}(p_b - p_s)$ and $c_{a \in A} = (e_a^c - e_a^g)p_s$. Furthermore, if $E^{Tr} > 0$, then $p_{tr} = p_s$.

Proof. We prove the first part of the lemma. The proof of the second part is analogous.

Assume that there is some non-critical net consumer d . Lemma A1 implies that $v_d = 0$. Since $d \in NC$, we have that

$$v_d = c_d^0 - c_d = \frac{e_a^{NC}}{\sum_{i \in NC} e_i^{NC}} E^{Tr} (p_b - p_{tr}) = 0$$

This implies that $E^{Tr} = 0$ or $p_{tr} = p_b$. In either case, we obtain

$$c_a = \begin{cases} e_a^{NC} p_b & \text{If } a \in NC \\ 0 & \text{If } a \in NO = (e_a^c - e_a^g) p_b \\ -e_a^{NP} p_b & \text{If } a \in NP \end{cases}$$

□

Proof of Proposition 1. Suppose net consumer $a \in A$ is not critical in community A . Lemma A1 implies that $v_a = c_a^0 - c_a = 0$. Thus, if no net consumer is critical, then all savings are allocated to net producers.

To see how savings are distributed among net producers, consider a net producer $h \in NP$ and the coalition $\Omega = NC \cup \{h\}$. Applying P7, we obtain $\sum_{i \in \Omega} v_i = v_h \geq e_h^{NP}(p_b - p_s)$, which, together with condition $\sum_{i \in A} v_i = \sum_{i \in A} e_i^{NP}(p_b - p_s)$ leads to $v_h = e_h^{NP}(p_b - p_s)$. This is the same allocation given by the EP CAM. The proof if there are no critical net producers is analogous. □

Proof of Proposition 2. Consider a community with 4 participants (1, 2, 3 and 4) in which there are two net consumers with $e_1^{NC} = 1$ and $e_2^{NC} = M > 0$, and two net producers with $e_3^{NP} = 1$ and $e_4^{NP} = M + 1$. The community as a whole is a net producer with $E^g - E^c = 1$ and $E^{Tr} = M + 1$. Agent 3 is not critical since $e_3^{NP} \leq E^g - E^c$. By Lemma A2, under any price-based CAM satisfying P7 the cost for participant 4 is $c_4^{(1)} = -p_s(M + 1)$.

If participant 4 increases its consumption in 2 units (so $e_4^{NP} = M - 1$, $E^c - E^g = 1$ and $E^{Tr} = M$), then agent 1 is not critical since $e_1^{NC} \leq E^c - E^g$. By Lemma A2, under any price-based CAM satisfying P7, the new cost for participant 4 is $c_4^{(2)} = -p_b(M - 1)$. The cost increment for participant 4 is $c_4^{(2)} - c_4^{(1)} = p_b + p_s - M(p_b - p_s)$. Consequently, if $M > \frac{p_b + p_s}{p_b - p_s}$, participant 4 decreases its cost by increasing its consumption in 2 units, violating property P4. □

Proof of Proposition 3. Consider a community with 3 participants (1, 2 and 3) where $e_1^{NC} = 1$, $e_2^{NP} = 1$ and a third agent with $e_3^g = 1$. We study the transfer price p_{tr} under any price-based CAM satisfying P7 when e_3^c tends to 1 from below and from above.

Note that the third agent is not critical, since $e_3^g - e_3^c = E^g - E^c$, and $E^{Tr} = 1$. By Lemma A2, under any price-based CAM satisfying P7, we have that:

- If $e_3^c > 1$, then the third player is a Net Consumer, so $p_{tr} = p_b$.
- If $e_3^c < 1$, then the third player is a Net Producer, so $p_{tr} = p_s$.

Thus, we have that under any price-based CAM satisfying P7

$$\lim_{e_3^c \rightarrow 1^+} p_{tr} = p_b \neq p_s = \lim_{e_3^c \rightarrow 1^-} p_{tr}$$

Therefore, the cost for the first agent as a function of agent 3's consumption is discontinuous:

$$\lim_{e_3 \rightarrow 1^+} c_1 = e_1^{NC} p_b = p_b \neq p_s = e_1^{NC} p_s = \lim_{e_3 \rightarrow 1^-} c_1$$

□

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